

IDEAS · E1410 · TEACHING PLAN

Advanced Project Management in AI

Eight live sessions (16 contact hours) paired with interactive self-study. Each 120-minute class follows a **20% lecture / 40% case studies / 40% interactive discussion & Q&A** split, anchored on a running project — **MekongPay**, a Vietnamese e-wallet building a fraud-detection AI — plus real industry cases. The workbook students build each session becomes their final case study.

PROGRAMME

MSc AI · IDEAS

SESSIONS

8 · 2h each

CONTACT HOURS

16h live · Zoom

SELF-STUDY

~20h + final

ASSESSMENT

50 / 50 MCQ · case

INSTRUCTOR · SCHEDULE

Dr. Jan Erik Meidell · August 2026 · Tue & Thu · 20:00–22:00 ICT / 15:00–17:00 CEST · live on Zoom, cameras on

ACCESS TO THE COURSE SITE

FOR THE CLASS · NOT THE INSTRUCTOR CONSOLE

Open the interactive course

The full course — live-class links, interactive lessons, readings and the project workbook — lives online. Open the site, enter the class access password, then register your name once (a personal 6-digit code is generated for you). The direct link below carries the password for you.

COURSE SITE

janerikmeidell.com/ideas-e1410

CLASS ACCESS PASSWORD

ideas1410x7k2

DIRECT LINK (PASSWORD INCLUDED)

janerikmeidell.com/ideas-e1410/index.html?auth=ideas1410x7k2

Shared by the whole class. The instructor dashboard is separate and PIN-protected.

TEACHING APPROACH & TIME ALLOCATION

Every live session is built on a fixed balance — **20% lecture/theory, 40% case studies, 40% interactive discussion & Q&A** — so the theory is always practised, not just heard. Below, hours are broken down per session across **lecture, interactive discussion, and self-study**, with case studies forming the practical core of each class.

■ Lecture / theory · 20% ■ Case studies · 40% ■ Interactive discussion & Q&A · 40% ■ Self-study (guided)

SESSION	TOPIC	LECTURE	CASE STUDIES	DISCUSSION / Q&A	LIVE TOTAL	SELF-STUDY
S1	Foundations of AI Project Management	24 min	48 min	48 min	2.0 h	2.0 h
S2	Planning & Scheduling AI Projects	24 min	48 min	48 min	2.0 h	2.5 h
S3	Resource Management in AI Projects	24 min	48 min	48 min	2.0 h	2.5 h
S4	Risk Management in AI Projects	24 min	48 min	48 min	2.0 h	3.0 h
S5	Agile & Team Management in AI Projects	24 min	48 min	48 min	2.0 h	2.5 h
S6	Monitoring, QA & Case Studies	24 min	48 min	48 min	2.0 h	2.5 h
S7	Case-Study Workshop — AI in the Real World	24 min	48 min	48 min	2.0 h	3.0 h
S8	Review & Final Assignment Preparation	24 min	48 min	48 min	2.0 h	2.0 h
Σ	Totals	3.2 h	6.4 h	6.4 h	16 h	~20 h

Sessions 7–8 keep the same ~20/40/40 balance but shift the practical block to applied work: in S7 the "case studies" time is students presenting their own AI projects, and the "discussion" time is structured peer critique; in S8 it is a synthesis review and final-assignment assembly. Self-study covers the interactive lesson, the reading companion, the self-check quiz and the project workbook. The final written case study is additional independent work (~16 h).

DETAILED CURRICULUM · SESSION BY SESSION

Each session maps to the official course lessons (L1–L6), with guiding questions, the theoretical & practical content covered, and the real case studies analysed in class. Technical terms and case names are kept in English as used in industry.

Foundations of AI Project Management

L1.1 · Nature of AI Projects

L1.2 · PM Principles in AI Contexts

L1.3 · AI Project Success Criteria

■ Lecture 24 min ■ Cases 48 min ■ Discussion 48 min

■ Self-study 2.0 h

GUIDING QUESTIONS

- ? **What makes an AI project structurally different from a traditional IT project?**
- ? **Which core PM knowledge areas still apply — and how must they adapt for AI?**
- ? **How do we define "success" for an AI system beyond model accuracy?**

THEORETICAL & PRACTICAL CONTENT

- ▶ Nature of AI projects: probabilistic vs deterministic outputs; experimentation & data dependency; core components (data, algorithms, compute); the AI project lifecycle
- ▶ PM principles in AI contexts: adapting the PMBOK knowledge areas — scope, time, cost, quality, risk, stakeholders; waterfall vs agile vs hybrid
- ▶ Success criteria: technical performance vs business value; model accuracy vs deployment success; adoption, operational impact and responsible-AI compliance
- ▶ Practical: launch the running project — frame your own AI initiative and start Workbook §1 (Context)

REAL CASE STUDIES ANALYSED

Google diabetic-retinopathy deep-learning model — 100k+ retinal images (Gulshan et al., JAMA 2016)

RAND (2024): ~80% of AI projects fail — why they fail differently

Automated hiring-tool bias · credit-approval AI reproducing discrimination

City facial-recognition deployment — stakeholder conflict

Running project → MekongPay — a Vietnamese e-wallet fraud-detection brief (team of 5, 6-month deadline, ~180M transactions). Students adopt it or their own case and complete §1 Context.

Planning & Scheduling AI Projects

L2.1 · AI Project Lifecycle & Stages

L2.2 · Scope & Requirements Engineering

L2.3 · Scheduling & Timeline Development

■ Lecture 24 min ■ Cases 48 min ■ Discussion 48 min

■ Self-study 2.5 h

GUIDING QUESTIONS

- ? Which lifecycle framework (CRISP-DM, TDSP, MLOps) fits a given AI project, and why?
- ? How do we turn a fuzzy business problem into functional & non-functional requirements?
- ? How do we schedule experimental work and communicate its uncertainty to stakeholders?

THEORETICAL & PRACTICAL CONTENT

- ▶ AI project lifecycle & frameworks: CRISP-DM, Microsoft TDSP, MLOps (data → train → validate → deploy → monitor → retrain)
- ▶ Scope & requirements engineering: business-to-ML objective mapping; functional vs non-functional requirements; data requirements; WBS; scope creep & change control
- ▶ Scheduling: iterative milestone planning; Critical Path Method (CPM); Gantt charts; parallel vs sequential tasks; time buffers; hybrid/Scrum; tools (MS Project, Jira, Asana)
- ▶ Practical: draft a lifecycle & milestone schedule — Workbook §2 (Why-AI & data) and §3 (Plan)

REAL CASE STUDIES ANALYSED

[Fraud-detection lifecycle: problem definition → deployment

[Predictive maintenance — 3 years of sensor data per machine

[Scheduling MekongPay's 6-month build under uncertainty

[Customer-churn project with evolving scope

Running project → Choose a lifecycle framework for MekongPay and build a milestone schedule with buffers.

Resource Management in AI Projects

L3.1 · Human Resource Management

L3.2 · Technical Infrastructure & Tools

L3.3 · Budgeting & Cost Estimation


■ Lecture 24 min ■ Cases 48 min ■ Discussion 48 min Self-study 2.5 h

GUIDING QUESTIONS

- ? Who belongs on an AI team, and how do business and technical roles collaborate?
- ? What infrastructure decisions (cloud vs on-prem, CPU vs GPU) shape the budget?
- ? How do we estimate cost when the work is experimental and uncertain?

THEORETICAL & PRACTICAL CONTENT

- ▶ Human resources: PM & stakeholder roles; core team (data scientist, ML engineer, data engineer, domain expert, product owner); vendor/contract management (data ownership, IP, SLAs)
- ▶ Technical infrastructure: data pipelines; compute (CPU/GPU); ML frameworks (TensorFlow, PyTorch, Scikit-learn); cloud platforms (AWS, Azure, GCP); MLOps & version control
- ▶ Budgeting: five cost categories (data, infrastructure, compute, personnel, deployment); estimation techniques (analogous, bottom-up, parametric, scenario); buffers & contingency
- ▶ Practical: staff, tool and budget the project — Workbook §4 (Resources)

REAL CASE STUDIES ANALYSED

- (Domain-expert roles: physicians (diagnostics) · financial analysts (fraud) · engineers (predictive maintenance)
- (Cloud & framework choices: AWS / Azure / GCP; TensorFlow / PyTorch
- (Streaming-recommender compute costs at scale (Budgeting MekongPay's 5-person, 6-month effort

Running project → Define the team, infrastructure and a costed budget for MekongPay.

Risk Management in AI Projects

L4.1 · Risk Identification

L4.2 · Risk Analysis & Mitigation

L4.3 · Governance & Compliance


 Lecture 24 min  Cases 48 min  Discussion 48 min Self-study 3.0 h

GUIDING QUESTIONS

- ? What risk categories are unique to AI, and how do we surface them early?
- ? How do we prioritise AI risks and choose mitigation strategies?
- ? What governance & regulation must an AI project satisfy to deploy responsibly?

THEORETICAL & PRACTICAL CONTENT

- ▶ Risk identification: five AI risk categories (data, algorithmic, operational, lifecycle, regulatory/ethical); sources of algorithmic bias; model drift
- ▶ Analysis & mitigation: qualitative probability–impact matrix vs quantitative Monte Carlo, sensitivity analysis / tornado, EMV; bias mitigation (pre/in/post-processing); four responses — avoid, reduce, transfer, accept
- ▶ Governance: OECD AI Principles; EU AI Act (risk-based); GDPR; explainable AI (SHAP, LIME); accountability & audit trails; NIST AI RMF
- ▶ Practical: build a risk register with mitigations — Workbook §5 (Risk)

REAL CASE STUDIES ANALYSED

- (Discriminatory credit-scoring & biased hiring (O'Neil, *Weapons of Math Destruction*, 2016)
- (Medical-diagnostic AI bias — high impact / medium likelihood
- (Credit-risk model sensitivity to income-data quality
- (MekongPay fraud thresholds (<2% false alarms) & State Bank of Vietnam compliance

Running project → Sessions 1–4 are examined in the mid-term MCQ. Build MekongPay's risk register.

Agile & Team Management in AI Projects

L5.1 · Agile Methodologies

L5.2 · Scrum in AI Projects

L5.3 · Communication & Stakeholders


 Lecture 24 min  Cases 48 min  Discussion 48 min Self-study 2.5 h

GUIDING QUESTIONS

- ? **Why does AI development favour agile / hybrid over pure waterfall?**
- ? **How does Scrum adapt when "done" means the experiment produced insight?**
- ? **How do we communicate model uncertainty to non-technical stakeholders?**

THEORETICAL & PRACTICAL CONTENT

- ▶ Agile methodologies: waterfall vs agile; the iterative experimentation loop; Lean AI; MVP / MVM; hybrid (structured governance + agile model development)
- ▶ Scrum for AI: roles (Product Owner, Scrum Master, dev team); sprints, backlog, sprint review by model metrics; validated learning; managing uncertainty
- ▶ Communication: probabilistic vs deterministic framing; translating precision/recall for executives; stakeholder mapping (power-interest); model cards & datasheets; conflict resolution
- ▶ Practical: sprint plan & stakeholder map — Workbook §6 (Delivery & monitoring)

REAL CASE STUDIES ANALYSED

- [Fraud-detection precision/recall explained to an operations team
- [Churn insight to executives: "30% of churn spotted two months earlier"
- [Hybrid PM structure for an AI rollout · sprint-review reframing
- [MekongPay: sprinting the fraud model & briefing the COO sponsor

Running project → Draft a sprint plan and stakeholder map for MekongPay.

Monitoring, Quality Assurance & Case Studies

L6.1 · Monitoring & Controlling

L6.2 · Quality Assurance

L6.3 · Case Studies of Implementation

■ Lecture 24 min ■ Cases 48 min ■ Discussion 48 min

■ Self-study 2.5 h

GUIDING QUESTIONS

- ? **How do we detect and respond to model drift after deployment?**
- ? **What does quality assurance mean for a probabilistic system?**
- ? **What separates AI projects that scale from those that stall?**

THEORETICAL & PRACTICAL CONTENT

- ▶ Monitoring & controlling: KPI categories (model, operational, business); drift types (data, concept, prediction); detection (PSI, KL divergence, thresholds); continuous-improvement loop
- ▶ Quality assurance: AI QA vs traditional; validation (train-test, holdout, K-fold cross-validation); over/underfitting; A/B testing; reproducibility & experiment tracking; documentation standards
- ▶ Case-study analysis through the PM lens: scope, stakeholders, risks, timeline & budget, lessons
- ▶ Practical: monitoring & QA plan — Workbook §7 (Responsible-AI) and §8 (Success & reflection)

REAL CASE STUDIES ANALYSED

- (Colgate-Palmolive — generative AI for concept development (MIT SMR, 2025)
- (Sanofi — AI decision support for investment choices (MIT SMR, 2025)
- (Liberty Mutual — AI-assisted claims triage (MIT SMR, 2025)
- (Toyota — 10,000+ man-hours/yr saved · BMW — Vertex AI fleet optimisation (Google Cloud, 2024)
- (Monitoring tools: MLflow · SageMaker Model Monitor · Vertex AI

Running project → Add a monitoring, drift-response and QA plan to MekongPay.

Case-Study Workshop — AI Projects in the Real World

Integrative · applies L1–L6

Student presentations + peer critique



■ Framing 24 min ■ Presentations 48 min ■ Peer critique 48 min

■ Self-study 3.0 h

GUIDING QUESTIONS

- ? **Does your project's plan survive scrutiny across scope, resources, risk and delivery?**
- ? **Where are the weakest assumptions, and how would you de-risk them?**

THEORETICAL & PRACTICAL CONTENT

- ▶ Students present the running AI project they have built across the workbook (§1–§8)
- ▶ Structured peer critique using the PM-lens dimensions: scope, stakeholders, risks, timeline & budget, responsible-AI
- ▶ Instructor feedback; identify and de-risk the weakest assumptions before the final submission
- ▶ Practical: consolidate everything in "Compile my project"

CASE MATERIAL

- (Each student's own AI project brief (MekongPay-style)
- (Cross-comparison of cohort approaches to the same constraints

Running project → Present, critique and refine — the case is now ready to write up.

Review & Final Assignment Preparation

Synthesis · L1–L6

Governance capstone + exam readiness


■ Synthesis 24 min ■ Case review 48 min ■ Discussion / Q&A 48 min Self-study 2.0 h + final

GUIDING QUESTIONS

- ? How do the eight knowledge areas connect into one coherent AI delivery approach?
- ? Is your final case study complete, critical and well-evidenced?

THEORETICAL & PRACTICAL CONTENT

- ▶ Course-wide synthesis: foundations → planning → resources → risk → agile/teams → monitoring/QA
- ▶ Responsible-AI & governance capstone pass over every project
- ▶ Mid-term debrief; final case-study assembly — structure, rubric, evidence; exam readiness
- ▶ Practical: finalise the written case study from the workbook

CASE MATERIAL

 Comparative review of the cohort's projects The term's marquee cases revisited (Google health AI · Colgate / Sanofi · MekongPay)

Running project → Submit the finished case study — your 50% final assignment.

OVERVIEW OF FINAL ASSIGNMENTS

50% · Mid-term — quiz exam

A **60-question multiple-choice exam** on the IDEAS / UMEF LMS, covering **Sessions 1–4**: the nature of AI projects, planning & scheduling, resource management, and risk. Delivered as an interactive quiz.

- ▶ Tests the fundamental concepts of AI project management.
- ▶ The self-check quiz in each session is the best rehearsal — same style, instant explanations.

50% · Final — written case study

The recommended route develops the **running project** — the one built in each session's workbook — into an in-depth written case study. Its eight sections map directly to the workbook; the final asks students to expand each into researched, critical analysis.

- ▶ **Sections:** §1 Context · §2 Why-AI & data · §3 Plan · §4 Resources · §5 Risk · §6 Delivery/monitoring · §7 Responsible-AI · §8 Success/reflection.
- ▶ **Rubric:** critical thinking 30 · research 25 · application 25 · structure 20.
- ▶ Alternative routes — an in-depth topic analysis or research paper — remain open.

COURSE LEARNING OUTCOMES

By the end of the course, students will be able to:

- ▶ Understand the principles of project management in **AI-specific** contexts.
- ▶ Apply advanced project-management techniques to AI projects.
- ▶ Manage resources and mitigate the risks tied to AI deployments.
- ▶ Develop effective timelines and budgets for AI projects.
- ▶ Evaluate AI project outcomes and ensure successful delivery.

MATERIALS, TOOLS & BIBLIOGRAPHY

Materials & delivery

- ▶ **Text:** Kai-Fu Lee, *AI Superpowers: China, Silicon Valley, and the New World Order* (2018) — with a reading companion.
- ▶ **Articles:** AI & project management (empirical) · Impact of AI on project management (PM2030) · AI adoption in SMEs & large firms — each with an interactive companion + PDF.
- ▶ **Slides:** an in-browser lecture deck per session, exportable to PDF.
- ▶ **Localised:** examples centred on a Vietnamese context (MekongPay), with an EN / VI interface.

Frameworks & standards referenced

- ▶ PMI PMBOK Guide (7th ed.) · CRISP-DM · Microsoft TDSP · MLOps
- ▶ NIST AI Risk Management Framework (AI RMF 1.0) · OECD AI Principles
- ▶ EU AI Act (risk-based) · GDPR · explainable AI (SHAP, LIME)
- ▶ Tooling: TensorFlow · PyTorch · Scikit-learn · AWS / Azure / GCP · MLflow · Jira

← **Program briefing** · **Live course**

© 2026 IDEAS · E1410 Advanced Project Management in AI · Dr. Jan Erik Meidell